

Chapter 2.

Moving averages and seasonal effects.

Situation

During 2014 the owners of a particular tourist attraction decide to use the data regarding the number of visitors the centre has had in recent years to identify trends and to indicate likely visitor figures for the future. The owners have access to the four monthly visitor totals for 2011, 2012 and 2013 and also for the first four months of 2014. These figures are shown in the table below.

Year	Period of time	Visitors (thousands)
2011	During 1st 4 months	45
2011	During 2nd 4 months	26
2011	During 3rd 4 months	34
2012	During 1st 4 months	52
2012	During 2nd 4 months	37
2012	During 3rd 4 months	42
2013	During 1st 4 months	64
2013	During 2nd 4 months	41
2013	During 3rd 4 months	50
2014	During 1st 4 months	69

- Whilst these visitor figures seem to jump around somewhat is the underlying trend one of
 - increasing numbers of visitors,
 - decreasing numbers of visitors,
 - or a reasonably constant number of visitors?
- If the next figure, i.e. for the 2nd period of 4 months of 2014, turns out to be 58 this is clearly a drop from the 69 recorded for the 1st four months of 2014 but is it cause for concern?

The situation on the previous page asked you to do two things:

1. Attempt to identify an underlying trend in data that may initially appear rather erratic.
2. Think about the significance of a value beyond those initially given.

In this chapter we will consider • making underlying trends more apparent,
and • making predictions,
for the sort of fluctuating data given on the previous page.

Making underlying trends more apparent.

The graph on the right shows the data from the situation on the previous page.

The graph shows the **seasonal nature** of the data with three "seasons" making up one cycle.

The peaks occur at the 1st, 4th, 7th and 10th points plotted and the troughs at the 2nd, 5th and 8th points plotted. From this pattern we would certainly expect the next value, i.e. for the 2nd 4 month period of 2014, to be a trough. A figure of 58 for this 4 month period would therefore not automatically be a cause for concern unless the drop were more than was expected.

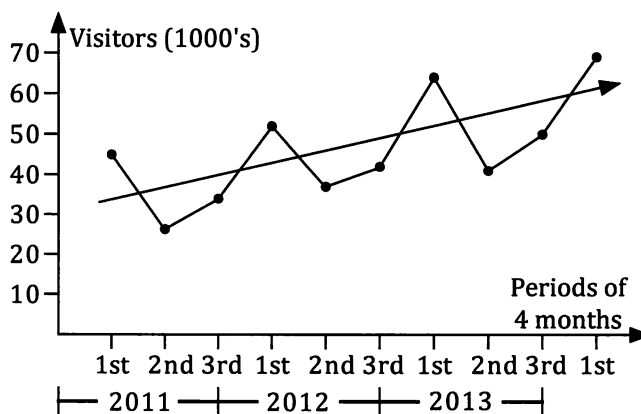
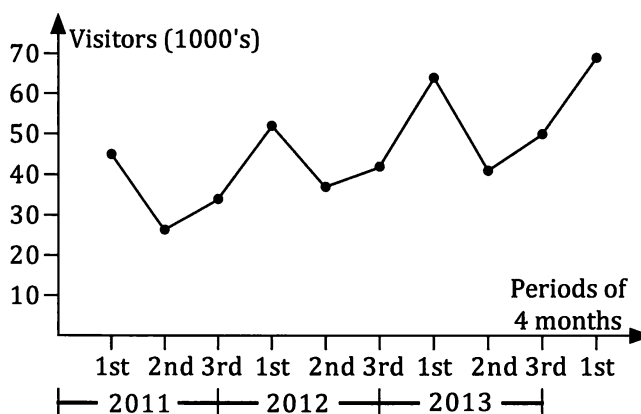
(Note: Seasonal variation has a constant time period. A repeating pattern without a regular frequency of occurrence would be called cyclical rather than seasonal.)

The graph indicates the general upward trend in the data as shown by the single line on the right.

The placement of this line is just a "guesstimate" based on "eyeballing" the graph and guessing an approximate line of best fit.

We could obtain a line of best fit using the linear regression techniques that you have encountered previously but, with the original data quite widely spread, and with an obvious seasonal nature, it is better to first *smooth* the graph by attempting to remove the seasonal variation.

We could smooth the data by taking an average attendance for each of the three years but this will reduce 10 data points down to just 3. Instead, because the given data has a three point pattern to it we create a **3 point moving average**.



Moving averages.

A 3 point moving average is found by first averaging the number of visitors for the

1st, 2nd and 3rd data points,

then averaging the

2nd, 3rd and 4th data points,

then averaging the

3rd, 4th and 5th data points, etc.

In this way we move down the data points and each average will be formed using a 1st 4 months figure, a 2nd 4 months figure, and a 3rd 4 months figure.

The first three 3 point moving averages are shown below (rounded to 1 decimal place when appropriate). Note carefully the placement of the moving average (MA) figures.

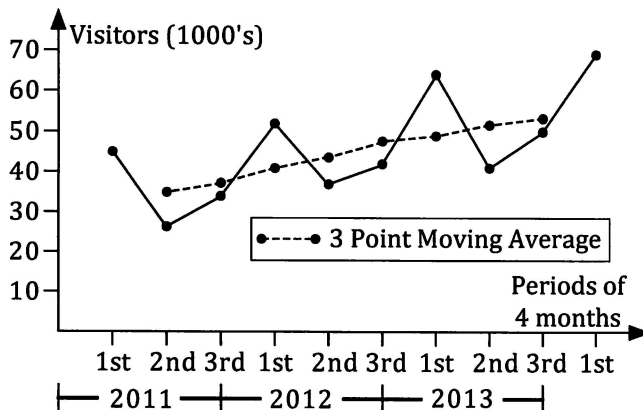
Yr	Months	MA
11	1st 4	45
	2nd 4	26 → 35
	3rd 4	34
12	1st 4	52
	2nd 4	37
	3rd 4	42

Yr	Months	MA
11	1st 4	45
	2nd 4	26
	3rd 4	34 → 37.3
12	1st 4	52
	2nd 4	37
	3rd 4	42

Yr	Months	MA
11	1st 4	45
	2nd 4	26
	3rd 4	34
12	1st 4	52 → 41
	2nd 4	37
	3rd 4	42

The complete table is shown on the right.

Adding these values to the graph (see the dashed line in the graph below) we see that this moving average technique has smoothed the data considerably.



Year	Months	Visitors	3pt MA
2011	1st 4	45	—
	2nd 4	26	35
	3rd 4	34	37.3
2012	1st 4	52	41
	2nd 4	37	43.7
	3rd 4	42	47.7
2013	1st 4	64	49
	2nd 4	41	51.7
	3rd 4	50	53.3
2014	1st 4	69	—

The moving average has smoothed the seasonal component in the data, leaving the trend component more apparent.

The three point moving average achieved this smoothing because the original data had a three point pattern to it. Had the original data had a four point pattern, as we might expect if we were collecting data every quarter year, we would use a four point moving average as shown in example 1 on the next page. If data was collected each month we might expect a 12 point pattern. If data was collected every day we might expect a 7 point pattern.

Example 1

To monitor the planned eradication of a particular disease in cattle all vets were required to notify the government of any cattle diagnosed as having the disease. The data collected was totalled every three months (or quarter of a year) and is shown on the right.

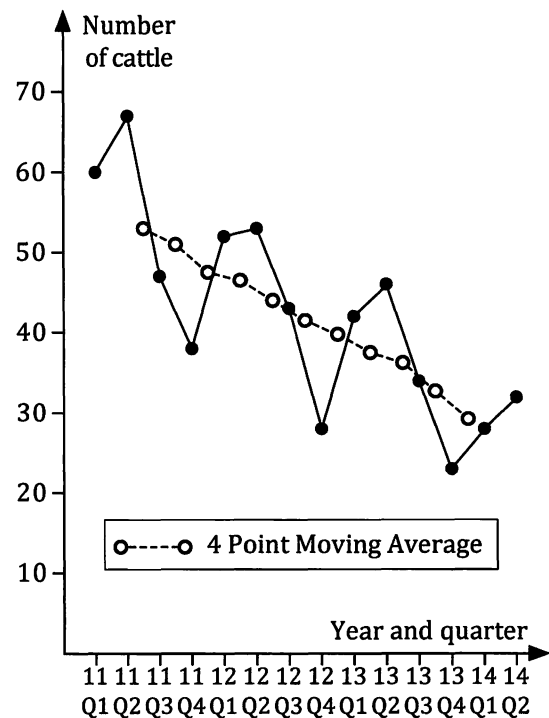
- Calculate the 4 point moving averages.
- On a single graph plot both the original data and the four point moving averages.
- Is the underlying trend decreasing, steady or increasing?

Year	Period of time	Number of cattle
2011	1st quarter	60
	2nd quarter	67
	3rd quarter	47
	4th quarter	38
2012	1st quarter	52
	2nd quarter	53
	3rd quarter	43
	4th quarter	28
2013	1st quarter	42
	2nd quarter	46
	3rd quarter	34
	4th quarter	23
2014	1st quarter	28
	2nd quarter	32

The 4 point moving averages are shown in the table below left. Again note carefully the placement of these figures.

The initial data and the 4 point moving averages are shown graphed below right.

Year	Period of time	No. of cattle	4 pt M A
2011	1st quarter	60	
	2nd quarter	67	
	3rd quarter	47	→ 53
	4th quarter	38	51
2012	1st quarter	52	47.5
	2nd quarter	53	46.5
	3rd quarter	43	44
	4th quarter	28	41.5
2013	1st quarter	42	39.75
	2nd quarter	46	37.5
	3rd quarter	34	36.25
	4th quarter	23	32.75
2014	1st quarter	28	29.25
	2nd quarter	32	

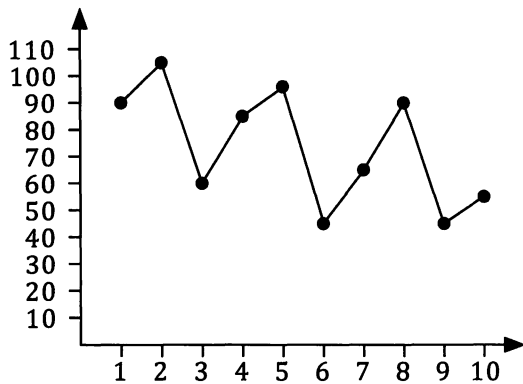


- The number of cattle diagnosed with the disease is showing a decreasing trend.

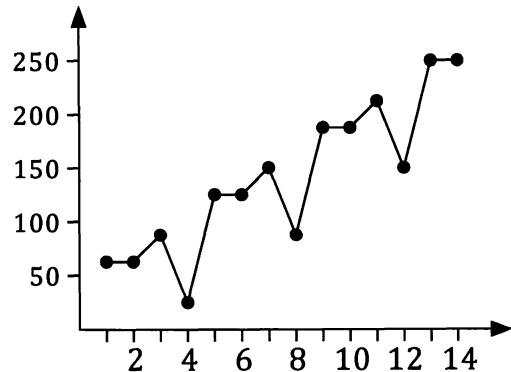
Exercise 2A

For each of numbers 1 to 6 suggest the most appropriate moving average to use to smooth the given set of data. (i.e. 3 point, 4 point, ? point.)

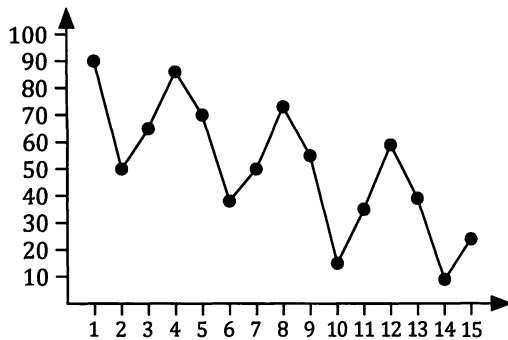
1.



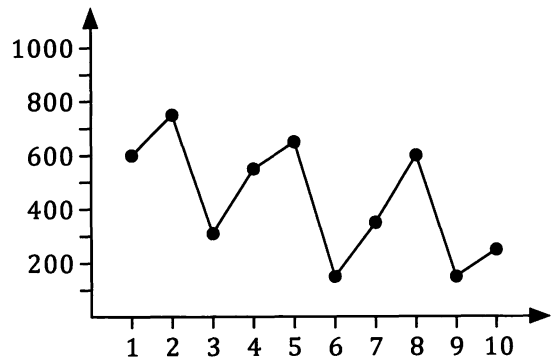
2.



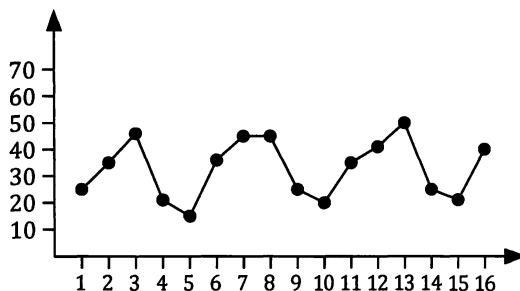
3.



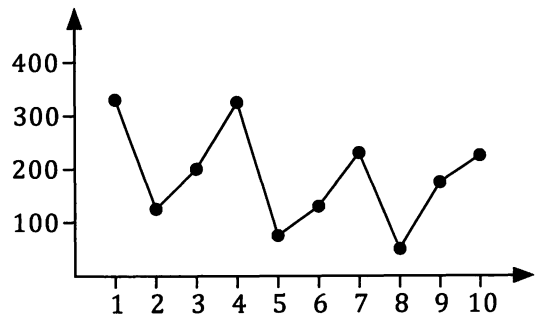
4.



5.



6.

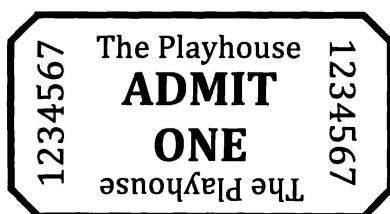


Note for the remaining questions of this exercise:

Some calculators and computer programs, when given the original data, can be programmed to list moving averages directly. However, even if you have access to such capability, do the remaining questions of this exercise without using this facility, to ensure that you fully understand the process.

7. A drama group put on a particular play three times a week for five weeks. The capacity of the theatre they use for the production is 500 and the numbers attending over the five week "season" are given below right.

- (a) Plot this data as a graph with the performance number on the horizontal axis and attendance on the vertical axis.
- (b) Add to your graph a plot of the data after it has been smoothed using a three point moving average.
- (c) Do the attendance figures indicate that a sixth week of performances would have been a good idea?

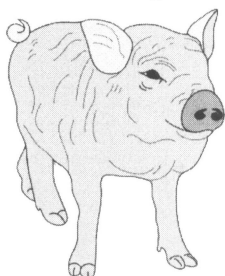


Performance Number	Week	Day	Attendance
1	1	Thur	210
2		Fri	320
3		Sat	370
4	2	Thur	270
5		Fri	440
6		Sat	450
7	3	Thur	420
8		Fri	500
9		Sat	500
10	4	Thur	450
11		Fri	500
12		Sat	500
13	5	Thur	350
14		Fri	410
15		Sat	430

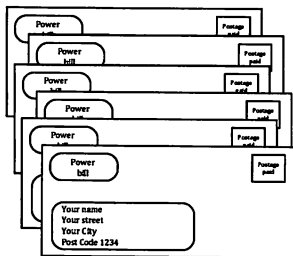
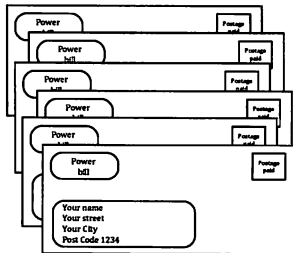
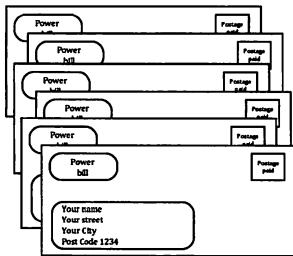
8. To monitor the planned eradication of a particular disease in pigs all vets were required to notify the government of any pig diagnosed as having the disease. The data collected was totalled every three months (or quarter of a year) and is shown on the right.

- (a) Calculate the 4 point moving averages.
- (b) On a single graph plot both the original data and the four point moving averages.
- (c) Is the underlying trend decreasing, steady or increasing?

Year	Period of time	Number of pigs
1	1st quarter	29
	2nd quarter	31
	3rd quarter	21
	4th quarter	25
2	1st quarter	28
	2nd quarter	24
	3rd quarter	21
	4th quarter	23
3	1st quarter	26
	2nd quarter	24
	3rd quarter	19
	4th quarter	20
4	1st quarter	25
	2nd quarter	19



9. The table below relates to the power bills received by a household every two months for a period of three years.



	Year	2 month period	Bill (\$)	6pt MA
1	2012	1st 2 months	238	—
2		2nd 2 months	251	—
3		3rd 2 months	283	258
4		4th 2 months	306	261
5		5th 2 months	274	A
6		6th 2 months	196	B
7	2013	1st 2 months	256	C
8		2nd 2 months	263	D
9		3rd 2 months	325	284
10		4th 2 months	342	287
11		5th 2 months	280	288
12		6th 2 months	238	287
13	2014	1st 2 months	274	289
14		2nd 2 months	269	290
15		3rd 2 months	319	293
16		4th 2 months	354	—
17		5th 2 months	286	—
18		6th 2 months	E	

- Determine values for the spaces marked A, B, C, D and E.
- On a single graph plot both the bill amounts for the three years and the six point moving averages.
- Is the underlying trend decreasing, steady or increasing?

General smoothing of time series data.

Even if there is no obvious seasonal nature to time series data we can still use a moving average to smooth out any **irregular fluctuations** and **unsystematic peaks and troughs**. For example consider the following table showing the total wool production in Australia, in tonnes, every five years from 1915 to 2010.

Year	1915	1920	1925	1930	1935
Total production (tonnes)	291300	300800	330700	425600	460500

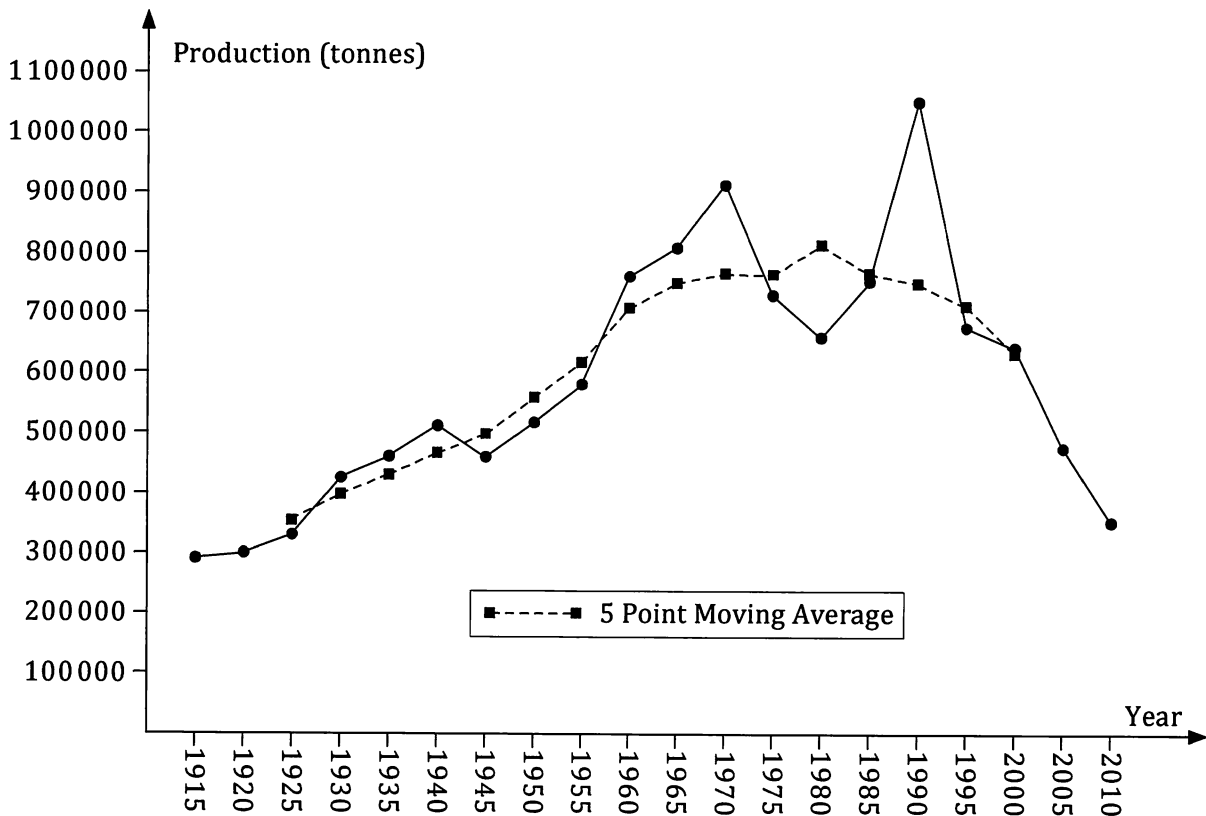
Year	1940	1945	1950	1955	1960
Total production (tonnes)	511400	460000	516800	580500	760500

Year	1965	1970	1975	1980	1985
Total production (tonnes)	807900	912100	729200	658200	752200

Year	1990	1995	2000	2005	2010
Total production (tonnes)	1049800	679400	641500	475200	352700

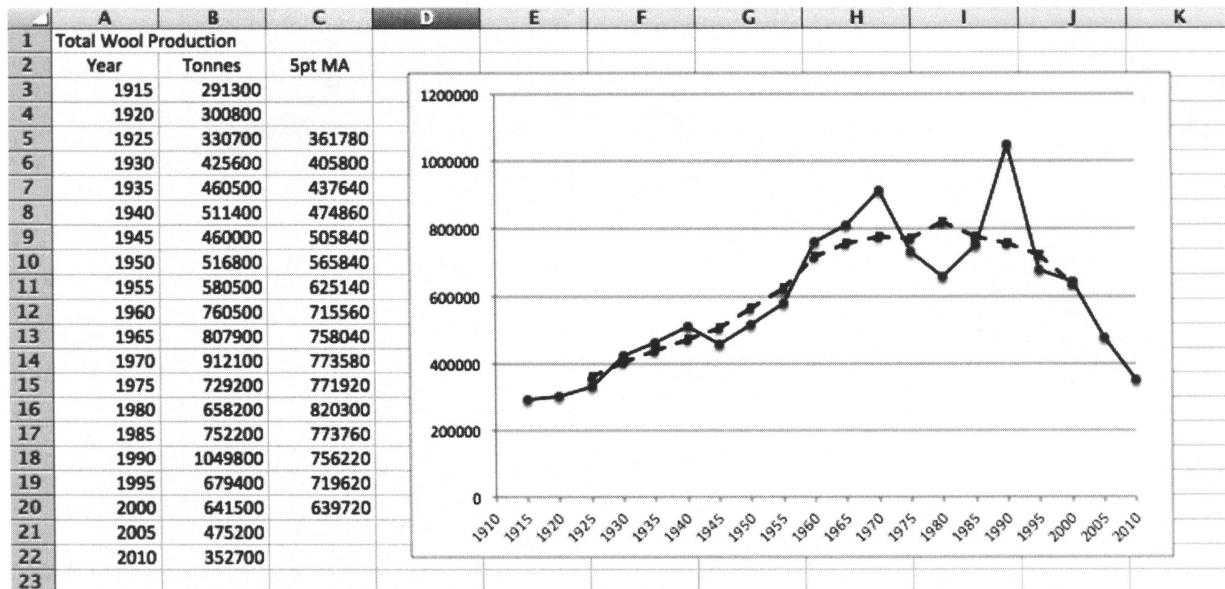
(Based on Australian Bureau of Statistics data.)

Plotting this data below shows no particular systematic, calendar related movements but plotting the moving average data (shown dashed) still has a smoothing effect and can make an underlying trend clearer.



Use of spreadsheets.

Spreadsheets can be particularly suitable for calculating moving averages and displaying the results graphically. The information from the previous situation involving wool production in Australia can be put into a spreadsheet, the moving averages can be calculated and the data displayed graphically, as shown below.



Try to create your own spreadsheets to answer some of the questions in this chapter.

Note: You may find that a spreadsheet offers the facility of adding a trend line to a graph, and one of the trend line options may be a moving average line for which you can set the period, i.e. a 3 point, 4 point, ... N point moving average. However it may be the case that the moving average points are plotted at the end of the period not at the centre, as we tend to show them. Plotting values at the end of the values that are used to form the moving averages produces the same line as shown dashed above, but moved to the right. In such a line the same values are displayed "later". The line is said to be showing the *trailing* moving averages. In this text we will not use this line and will instead always plot our moving averages at the centre of the time values used to form it, as shown above.

However, that is not to say that attaching a moving average to the end of the period used to form it cannot be useful in some circumstances. Suppose for example that you were an athlete monitoring your times in training. Today you might well form a three point average involving today's time and the times of the previous two days. In your training diary you could well record today's figure as being the average of today's time and that of the previous two days. I.e. you would be recording the three point moving average at the end of each three day period used to calculate it. But, as stated above, when plotting moving averages in this text in order to smooth data we will plot at the centre of the time interval used to calculate it.

Centred moving averages.

Consider again the calculation of four point moving averages encountered previously involving data collected with regard to a disease in cattle, as shown on the right. Note the moving average figure does not align with a particular quarter but instead falls between quarters.

This "misalignment" of original data and moving average figures will occur in an N point moving average when N is even. For some uses of moving averages it is more convenient if the two quantities do align. Hence it is quite common to give an N point moving average *centred* when N is even. This centring simply involves taking the average of two values, as the following table shows for the cattle disease data.

Year	Quarter	No. of cattle	4 pt M A
2011	1st	60	
	2nd	67	
	3rd	47	→ 53
	4th	38	

<i>t</i>	Year	Quarter	No. of cattle	4 pt MA	Centred 4 pt MA
1	2011	1st	60		
2		2nd	67		
3		3rd	47	→ 53	→ 52
4		4th	38	51	49.25
				47.5	
5	2012	1st	52	46.5	47
6		2nd	53	44	45.25
7		3rd	43	41.5	42.75
8		4th	28	39.75	40.625
9	2013	1st	42	37.5	38.625
10		2nd	46	36.25	36.875
11		3rd	34	32.75	34.5
12		4th	23	29.25	31
13	2014	1st	28		
14		2nd	32		

Note: • We can go directly from the initial data to the centred four point moving average if we take the **five** values that contribute to the centred average and sum "half the first, half the last, and the middle three" and then divide by four. Applying this idea to the previous example, the first five values give

$$\frac{30 \text{ (i.e. half of 60)} + 67 + 47 + 38 + 26 \text{ (i.e. half of 52)}}{4} = 52, \text{ as required.}$$

The next set of five values give:

$$\frac{33 \cdot 5 + 47 + 38 + 52 + 26 \cdot 5}{4} = 49 \cdot 25, \text{ as required.}$$

- ☞ Use a, b, c, d and e to prove that the above method gives the same result as is obtained by first obtaining the uncentred 4 point averages.
- ☞ What would be the rule for centring six point averages in this way?

Exercise 2B

1. The table below shows the number of cans of soft drink a lunch bar sells in each three months of a three year period.
 - (a) Why would determining 4 point moving averages be suitable for this data?
 - (b) Determine the values A, B, C, D, E, F and G.

<i>t</i>	Year	Quarter	Cans sold	4pt MA	Centred 4pt MA
1	One	1st	8 312	—	—
2		2nd	5 783	—	—
3		3rd	5 727	7 006	6 983
4		4th	8 202	6 960	A
				6 920	
5	Two	1st	8 128		6 906
				B	
6		2nd	5 623		6 840
7		3rd	5 615	C	D
8		4th	7 786	6 738	
				6 710	
9	Three	1st	7 928		6 698
				6 686	
10		2nd	E		G
11		3rd	5 519	F	—
12		4th	7 594	—	—

2. The table below shows the number of client interviews an accountant conducts in each two months for a period of three years.

(a) Determine the values A, B, C, D E, F, G, H, I and J.

<i>t</i>	Year	2 month period	Units	6pt MA	Cent'd 6pt MA
1	1	1st	142		
2		2nd	235		
3		3rd	295		
4		4th	141	175	
5		5th	80	178	A
6		6th	157	B	C
7	2	1st	160	D	
8		2nd	247	184	
9		3rd	319	185	183.5
10		4th	135	E	F
11		5th	92	185	
12		6th	163	186	185.5
13	3	1st	148	184	
14		2nd	217	179	181.5
15		3rd	343	183	181
16		4th	165	188	
17		5th	104	H	I
18		6th	G	193	J

- (b) Reproduce the centred 6 point moving average figures using a spreadsheet showing only two columns, the raw data "units" column (with your value for G included) and the centred moving average column. (See the note at the top of the previous page.)

3. Determine the moving averages for the information given below.
(The information shows the tonnage of a particular crop produced in each season, over two years. Take MAs to mean Moving Averages and CMAs to mean Centred Moving Averages.)

Year 1				Year 2			
Summer	Autumn	Winter	Spring	Summer	Autumn	Winter	Spring
1228	364	640	1220	1436	276	752	1132

4 Point MAs

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4 Point CMAs

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4. The monthly average price per kilogram of a particular type of fruit was monitored over a period of four years. The data obtained is shown below.

Year 1					
January	February	March	April	May	June
\$2.90	\$2.70	\$1.90	\$1.80	\$1.75	\$1.90
July	August	September	October	November	December
\$1.90	\$1.95	\$2.00	\$2.15	\$2.60	\$3.00

Year 2					
January	February	March	April	May	June
\$3.30	\$2.40	\$1.90	\$1.95	\$2.00	\$1.95
July	August	September	October	November	December
\$2.00	\$2.10	\$2.25	\$2.35	\$2.60	\$2.80

Year 3					
January	February	March	April	May	June
\$3.20	\$3.60	\$2.60	\$2.20	\$2.05	\$2.00
July	August	September	October	November	December
\$2.00	\$2.05	\$2.10	\$2.20	\$2.60	\$3.10

Year 4					
January	February	March	April	May	June
\$3.20	\$2.65	\$2.05	\$1.95	\$2.05	\$2.20
July	August	September	October	November	December
\$2.40	\$2.50	\$2.60	\$2.90	\$3.40	\$4.30

- Why does it make sense to consider the 12 point moving averages for this data?
- Either manually, or with the assistance of a graphic calculator, spreadsheet or computer program display the original data, and the 12 point centred moving averages, on a single graph.
- Is the underlying trend decreasing, steady or increasing?

5. If the population of a region increases we might expect the number of criminal incidents reported by the police for that region to similarly increase. For crimes reported by New South Wales police and classified as "*Robbery with a firearm*" the numbers reported annually from 1990 to 2013 were as follows (rounded to the nearest multiple of 5):

Number of incidents classified as "Robbery with a firearm"
recorded by NSW police, 1990 to 2013.

Year	Number
1990	980
1991	1265
1992	1130
1993	875
1994	520
1995	785
1996	1015
1997	1400

Year	Number
1998	1120
1999	890
2000	805
2001	1105
2002	890
2003	835
2004	710
2005	520

Year	Number
2006	605
2007	595
2008	420
2009	520
2010	445
2011	400
2012	380
2013	325

(© State of New South Wales through the Department of Justice and reproduced with the approval of the NSW Bureau of Crime Statistics and Research.)

- Which would smooth the data more, a three point moving average or a five point moving average?
- Display on a single graph the original figures, and whichever of the three or five point moving averages you felt would smooth the data more.
- Comment on any trends shown.

Moving averages and finance.

Moving averages are sometimes used by people monitoring the movement of share prices. A 30 day moving average of share prices will be less influenced by a sudden change in the share price than say a 5 day moving average would be. By considering and comparing different moving averages a stock market analyst can attempt to identify trend and predict future movement in share prices. In some cases the analyst might "weight" the moving average so that the more recent share prices have more influence on the average than a price from a few weeks ago. Whilst this idea of a weighted moving average is beyond the scope of this course do a bit of research on the internet to see what some financial sites say about the use of moving averages.

Quantifying the seasonal effect.

Let us consider again the situation from the beginning of this chapter concerning visitor numbers to a particular tourist attraction. These figures are shown again below.

Year	Period of time	Visitors (thousands)
2011	During 1st 4 months	45
2011	During 2nd 4 months	26
2011	During 3rd 4 months	34
2012	During 1st 4 months	52
2012	During 2nd 4 months	37
2012	During 3rd 4 months	42
2013	During 1st 4 months	64
2013	During 2nd 4 months	41
2013	During 3rd 4 months	50
2014	During 1st 4 months	69

Notice that there seems to be a "1st 4 months effect" that leads to a higher number of visitors attending the tourist attraction in the first four months than in the other two periods of four months in each year.

Consider the figures for 2011:

$$\begin{aligned}
 \text{Total number of visitors in 2011} &= 45000 + 26000 + 34000 \\
 &= 105000 \\
 \text{Average number of visitors per 4 month period} &= 105000 \div 3 \\
 &= 35000
 \end{aligned}$$

The 1st 4 month number, 45000, as a percentage of this average figure, is 128.6%. (1dp)

The 2nd 4 month number, 26000, as a percentage of this average figure, is 74.3%. (1dp)

The 3rd 4 month number, 34000, as a percentage of this average figure, is 97.1%. (1 dp)

Hence we could say that for 2011 the 1st 4 months have a seasonal effect that lifts the attendance by 29% of the average, the 2nd 4 months have a seasonal effect that drops the attendance by 26% of the average (74% is 26% below 100%) and the 3rd 4 months have a seasonal effect that drops the attendance by 3% of the average.

However, with the figures for 2012 and 2013 available it would be better to carry out the above process for all three years and then average the results. This is indeed how we will determine a measure of the effect each season has – called the **seasonal effect**, **seasonal component** or **seasonal index**.

The three **seasonal indices** for the above data are calculated on the next page.

Year	Period of time	Visitors (1000s)	4 month mean for the year (1000s).	Visitor numbers as percentage of 4 month mean.
2011	1st 4 months	45	35	128.57%
	2nd 4 months	26		74.29%
	3rd 4 months	34		97.14%
2012	1st 4 months	52	43.667	119.08%
	2nd 4 months	37		84.73%
	3rd 4 months	42		96.18%
2013	1st 4 months	64	51.667	123.87%
	2nd 4 months	41		79.35%
	3rd 4 months	50		96.77%
2014	1st 4 months	69		

Calculation of seasonal effect, seasonal component or seasonal index:

	1st 4 months	2nd 4 months	3rd 4 months
2011	128.57%	74.29%	97.14%
2012	119.08%	84.73%	96.18%
2013	123.87%	79.35%	96.77%
Mean (nearest %)	124%	79%	97%

The seasonal index for the first 4 months of a year is 124% (or 1.24), the seasonal index for the second 4 months of a year is 79% (or 0.79) and the seasonal index for the third 4 months of a year is 97% (or 0.97). This means that visitor numbers for the first four months of a year tend to be 24% above the average 4 month numbers, for the second four months they tend to be 21% below the average 4 month numbers and for the third 4 months of a year they tend to be 3% below the average 4 month numbers.

Note • This particular method of determining the **seasonal indices** is called the **average percentage method**. There are a number of other methods that are used to quantify seasonal effects in time series data but in this unit we will use this average percentage method.

- The average of the seasonal indices should be 100%, i.e. with three seasons as shown above the total should be 300%. This is the case for the above indices, $124 + 79 + 97 = 300$.

If small errors due to rounding make an average not equal to 100 suitable adjustment could be made.

- On occasions, though rarely in this text, the average percentage method uses the median rather than the mean of the percentages. From the above table this would give indices of 123.87%, 79.35% and 96.77% (i.e. in this case it would make little difference, but that would not always be the case). In this text, when asked for the seasonal indices to be calculated using the average percentage method, the reader should assume the mean is to be used.

Read through the following which shows the calculation of seasonal indices for data involving six seasons and make sure you understand the calculations involved.

t	Year	2 month period	Units	2 month mean for the year.	Percentage of 2 month mean.
1	1	1st	284	352	80.68%
2		2nd	463		131.53%
3		3rd	595		169.03%
4		4th	279		79.26%
5		5th	168		47.73%
6		6th	323		91.76%
7	2	1st	318	370	85.95%
8		2nd	493		133.24%
9		3rd	635		171.62%
10		4th	268		72.43%
11		5th	188		50.81%
12		6th	318		85.95%
13	3	1st	302	381	79.27%
14		2nd	428		112.34%
15		3rd	678		177.95%
16		4th	308		80.84%
17		5th	211		55.38%
18		6th	359		94.23%

	1st 2 months	2nd 2 months	3rd 2 months	4th 2 months	5th 2 months	6th 2 months
1st year	80.68%	131.53%	169.03%	79.26%	47.73%	91.76%
2nd year	85.95%	133.24%	171.62%	72.43%	50.81%	85.95%
3rd year	79.27%	112.34%	177.95%	80.84%	55.38%	94.23%
Mean	82.0%	125.7%	172.9%	77.5%	51.3%	90.6%

The seasonal indices for the 1st 2 months, 2nd 2 months etc are

1st 2 months: 82.0% (or 0.820), 2nd 2 months: 125.7% (or 1.257),
 3rd 2 months: 172.9% (or 1.729), 4th 2 months: 77.5% (or 0.775),
 5th 2 months: 51.3% (or 0.513) 6th 2 months: 90.6% (or 0.906).

Notice that $82.0 + 125.7 + 172.9 + 77.5 + 51.3 + 90.6 = 600$, as we would want for data involving six seasons as then the average index is 100. However, were we to round each index to an integer we would have indices of 82, 126, 173, 78, 51 and 91 which give a total of 601, the "extra 1" being introduced due to the rounding carried out. To avoid this situation we could choose to round the 77.5 down to 77, to give indices of 82, 126, 173, 77, 51 and 91. However in real applications we would probably be going on to use these seasonal indices for further calculations so we would be able to use more accurate values stored in the cell of a spreadsheet and would leave all rounding until the final calculation is completed.

Exercise 2C

- Three of the seasonal indices for data with a quarterly seasonal pattern are as follows:

Seasonal index for the first quarter:	113%
Seasonal index for the second quarter:	105%
Seasonal index for the third quarter:	84%

Determine the seasonal index for the fourth quarter.

- Six of the seasonal indices for data with a daily seasonal pattern are as follows:

Seasonal index for Monday:	73%
Seasonal index for Tuesday:	78%
Seasonal index for Thursday:	89%
Seasonal index for Friday:	102%
Seasonal index for Saturday:	143%
Seasonal index for Sunday:	121%

Determine the seasonal index for Wednesday.

- A company expects the annual sales of one of their products next year to be 156000 and, from past years' experience, anticipates no underlying increasing or decreasing trend, and no irregular fluctuations in sales through the year.

- If there is no monthly seasonal effect on sales of this product roughly how many of the product should the company expect to sell each month?
- In fact the sales are seasonal in nature with the seasonal indices for each month being as follows:

January	88%	February	82%	March	76%
April	74%	May	83%	June	98%
July	104%	August	105%	September	108%
October	111%	November	126%	December	145%

Predict the sales numbers for this product for each month of next year.

- A company finds that in the first quarter of the year it sells 17360 units of a particular product.

From the quarterly sales of this product in previous years the company has calculated the seasonal indices for sales of this product per quarter as being as follows:

Seasonal index for the first quarter:	112%
Seasonal index for the second quarter:	104%
Seasonal index for the third quarter:	98%
Seasonal index for the fourth quarter:	86%

Assuming that for this year the only anticipated change in sales from one month to the next is due to this seasonal effect, what do these figures suggest for the likely number of units of this product sold:

- in the whole year,
- in each of the remaining three quarters of the year?

Copy and complete each of the following sets of seasonal data, filling in the spaces marked "?". (Whilst the average of the indices should be 100% do not worry if, due to rounding, this is not quite the case.)

5. Week	Period of time	Attendance	Daily mean for the week.	Attendance as % of week's daily mean.
1	Friday	2346	2766	84.82%
	Saturday	3143		113.63%
	Sunday	2809		101.55%
2	Friday	2572	?	?
	Saturday	3258		?
	Sunday	2957		100.96%
3	Friday	2987	?	?
	Saturday	3500		?
	Sunday	3206		99.23%

	Friday	Saturday	Sunday
1st week	84.82%	113.63%	101.55%
2nd week	?	?	100.96%
3rd week	?	?	99.23%
Index (1 dp)	?	?	100.6%

6. Year	Period of time	Units sold	Quarterly mean for the year.	Units as % of year's quarterly mean.
1	1st quarter	54000	52650	102.56%
	2nd quarter	63200		120.04%
	3rd quarter	45200		85.85%
	4th quarter	48200		?
2	1st quarter	48500	?	102.65%
	2nd quarter	56700		?
	3rd quarter	41300		?
	4th quarter	42500		?
3	1st quarter	51800	?	104.28%
	2nd quarter	62600		?
	3rd quarter	40500		?
	4th quarter	43800		?

	1st quarter	2nd quarter	3rd quarter	4th quarter
1st year	102.56%	120.04%	85.85%	?
2nd year	102.65%	?	?	?
3rd year	104.28%	?	?	?
Seasonal Index (1 dp)	103.2%	?	?	?

7. A local council commences a campaign to encourage people who need to contact the council for various matters to do so via the council's interactive website, thus avoiding the delays experienced by people making phone or direct contact with the council's help desk.

To monitor the success of this venture in reducing the number of calls coming into the help desk, and to be able to predict future staffing needs, the council monitors the number of non internet phone calls made to the help desk on each day of a three week period. The results are shown in the table below.

The council wants to know appropriate moving average figures for the three week period and the seasonal index for each day.

With the assistance of a spreadsheet construct completed versions of the following tables.

n	Week	Day	Nº. of calls	5pt MA	Calls as % of daily average for week.
1	One	Mon	258	—	107.59%
2		Tue	231	—	96.33%
3		Wed	215	239.8	
4		Thur	248	235.6	
5		Fri	247	231.0	
6	Two	Mon	237		
7		Tue	208		
8		Wed	184		
9		Thur	232		
10		Fri	223		
11	Three	Mon	232		
12		Tue	194		
13		Wed	183		
14		Thur	209	—	
15		Fri	208	—	

	Week 1	Week 2	Week 3	Seasonal index
Monday	107.59%			
Tuesday	96.33%			
Wednesday				
Thursday				
Friday				

Deseasonalising, or seasonally adjusting, the data.

Now that we are able to quantify the effect a season has we can remove this seasonal effect and present the data *deseasonalised* or *seasonally adjusted*. Considering again the visitor numbers for the tourist attraction situation encountered earlier:

Year	Period of time	Visitor (1000s)
2011	During 1st 4 months	45
2011	During 2nd 4 months	26
2011	During 3rd 4 months	34
2012	During 1st 4 months	52
2012	During 2nd 4 months	37
2012	During 3rd 4 months	42
2013	During 1st 4 months	64
2013	During 2nd 4 months	41
2013	During 3rd 4 months	50
2014	During 1st 4 months	69

We calculated earlier that the seasonal index for the 1st 4 months is 124%, or 1.24. Thus the figure of 45000 visitors includes a 24% "seasonal increase". To remove this 24% increase we divide the 45000 by 1.24 to give 36 thousand, to the nearest thousand. The seasonal index for the 2nd 4 months is 79%, or 0.79. Thus the figure of 26000 visitors includes a 21% "seasonal decrease". To remove this 21% decrease we divide the 26000 by 0.79 to give 33000, to the nearest thousand.

Continuing in this way gives the seasonally adjusted figures shown in the table below:

Year	Period of time	Visitors (1000s)	Seasonally adjusted visitor numbers. (1000s)
2011	During 1st 4 months	45	36
2011	During 2nd 4 months	26	33
2011	During 3rd 4 months	34	35
2012	During 1st 4 months	52	42
2012	During 2nd 4 months	37	47
2012	During 3rd 4 months	42	43
2013	During 1st 4 months	64	52
2013	During 2nd 4 months	41	52
2013	During 3rd 4 months	50	52
2014	During 1st 4 months	69	56

Seasonally adjusted unemployment figures.

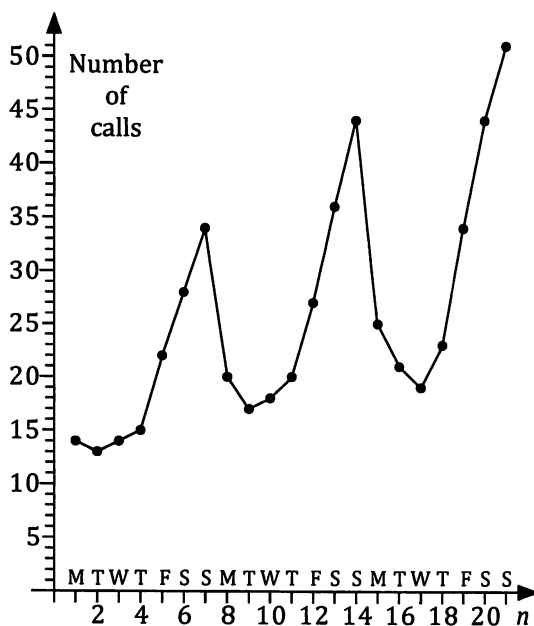
- Why might the seasons of the year cause the number of people employed in agriculture to vary?
- Why might the seasons of the year cause the number of people employed in hospitality to vary?
- Why might the number of people registered as unemployed rise at the end of a school year?
- Why is it common practice for government statistical agencies to quote unemployment rates as *seasonally adjusted* figures?

Making predictions.

Let us suppose that a bank decides to operate a *7-days a week phone help line* to give advice regarding new home loans and that the number of calls this help line receives in the first three weeks of operation are as shown in the table below.

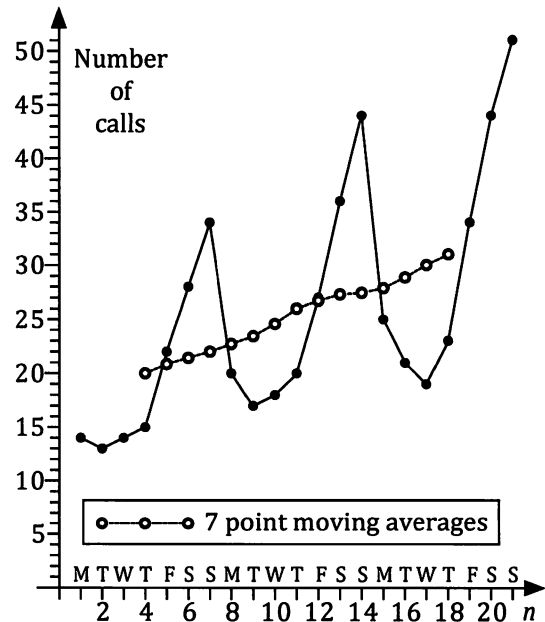
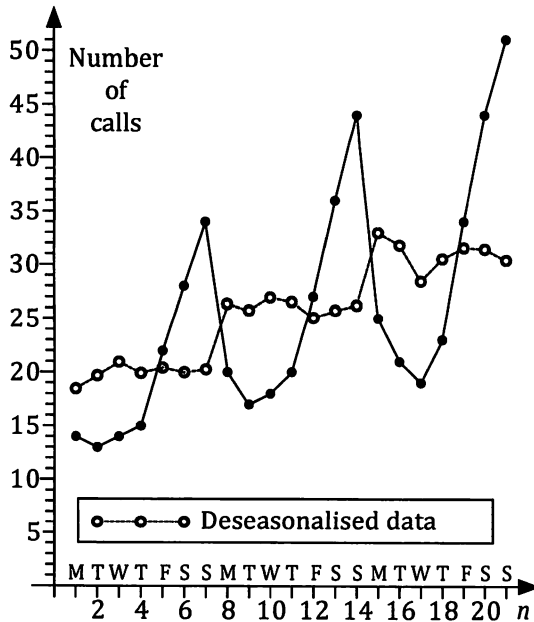
Can we use these figures to predict the number of calls this help line is likely to receive on the Monday and Saturday of week 4 (assuming any trends of the first three weeks continue)?

If we plot the data, as shown below, it might appear that any prediction would be difficult given the fluctuating nature of the points.



<i>n</i>	Week	Day	Nº. of calls
1	1	Mon	14
2		Tue	13
3		Wed	14
4		Thur	15
5		Fri	22
6		Sat	28
7		Sun	34
8	2	Mon	20
9		Tue	17
10		Wed	18
11		Thur	20
12		Fri	27
13		Sat	36
14		Sun	44
15	3	Mon	25
16		Tue	21
17		Wed	19
18		Thur	23
19		Fri	34
20		Sat	44
21		Sun	51

However, if we add the 7 point moving average data to this graph, or if we plot the deseasonalised data, both shown on the next page, we have straighter lines, showing less fluctuation. We could then, with more confidence, use linear regression techniques to predict future values (although we know that such prediction would involve extrapolation and so any predicted values would need to be viewed with caution).



- With each plotted point on the line of moving averages being the average of seven points it is perhaps to be expected that the moving average line shows less variation than the deseasonalised data line. One "unusual value" will have limited influence in changing the direction of a line when it is but one of seven scores given equal weighting when determining the moving average. Moving averages tend to remove seasonality and individuality. They smooth all variation, not just seasonal variation.
- The deseasonalised data was formed using the following seasonal indices calculated using the average percentage method explained earlier in the chapter.
Monday: 0.76 Tuesday: 0.66 Wednesday: 0.67 Thursday: 0.75
Friday: 1.08 Saturday: 1.40 Sunday: 1.68

Now if we are to use either the moving average values, or the deseasonalised values, to determine a line of best fit using linear regression techniques, any prediction we make using such a line of best fit will be a predicted moving average value, or a predicted deseasonalised value. For these predictions to be "real life values", rather than values that have been smoothed in some way, we will need to factor back in the seasonal effect.

For example, if we use linear regression to determine a line of best fit for predicting D , the deseasonalised data, given the value of n , the "day number", where $n = 1$ corresponds to Monday of week 1, we obtain the equation $D = 0.70n + 18.0$.

Thus: When $n = 22$, (i.e. Monday of week 4) predicted $D = 33.4$.

When $n = 27$, (i.e. Saturday of week 4) predicted $D = 36.9$.

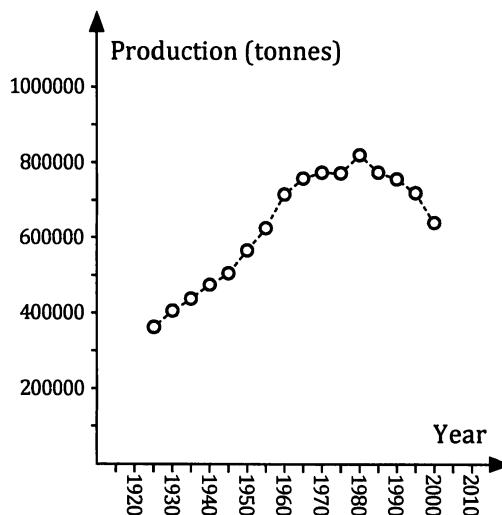
Hence to "factor back in the seasonal effect" the predictions will be as follows:

For Monday of week 4: Predicted number of calls = 33.4×0.76 , i.e. approx 25.

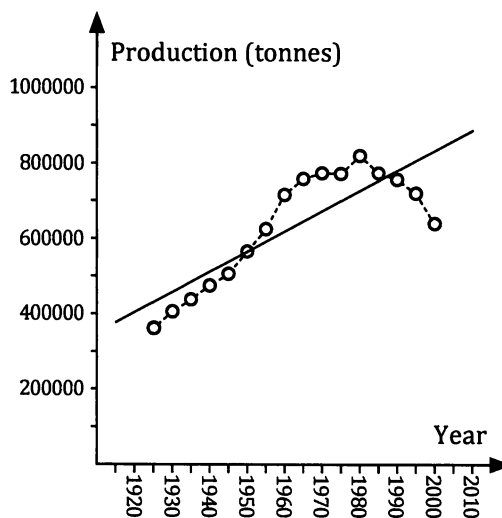
For Saturday of week 4: Predicted number of calls = 36.9×1.40 , i.e. approx 52.

Note: • Remember that not all data can be suitably modelled using linear regression. Just because smoothing techniques such as deseasonalising the data or finding the moving averages have been carried out we cannot assume that the resulting data is necessarily showing a linear pattern.

Recall for example the wool production data encountered earlier in this chapter, for which the moving average plot is shown below.



As the graph indicates, applying linear regression techniques to the whole data set would **not** be appropriate.



The straight line given by using linear regression is shown above and is not a very good fit for the data.

Remember:

$$\text{Deseasonalised data (or seasonally adjusted data)} = \frac{\text{Real data}}{\text{Seasonal index}}$$

and so

$$\text{Real data} = \text{Deseasonalised data} \times \text{Seasonal index}$$

Note: In this unit we are using a *multiplicative model* for determining the seasonal indices. Hence the above statements are true. In some case, though not in this unit, an additive model could be used and in such cases seasonal adjustment occurs by addition or subtraction, not by multiplication.

Exercise 2D

1. A company achieves profits of \$25400 in June of one year. If the June profit figures for this company have a seasonal index of 114% what would be the deseasonalised, or seasonally adjusted, profit for June? (Give your answer to the nearest \$100.)
2. From an analysis of sales from earlier years a company statistician determines that the number of units of a particular product the company sells follows a seasonal pattern with the following quarterly indices:

First quarter	0.65	Second quarter	0.76
Third quarter	0.82	Fourth quarter	1.77

 Deseasonalise a figure of 132000 units sold in the second quarter, giving your answer to the nearest 100 units.
3. (a) It is announced that a particular region had 65470 people unemployed in January "seasonally adjusted". If the seasonal index for January is 106.2% what was the "real number" unemployed for this region in January? (Give your answer to the nearest ten people.)
 (b) In May of the same year the region had 67140 people unemployed. If the seasonal index for May is 96% what would the real figure of 67140 be reported as after "seasonal adjustment"? (Again give your answer to the nearest ten people.)
4. Projecting forward using smoothed data a company analyst finds that the predicted sales figure for January will be 12500 units. However this has not allowed for the January seasonal effect. If the seasonal index for January is 1.18 what does this predict for the number of units sold in January?
5. Projecting forward using smoothed data the statistically astute owner of a number of orchards predicts that the orchards will produce 16000 kg of a particular fruit this summer, but he knows that this prediction has not yet allowed for the seasonal effect of summer. If the summer seasonal index is 1.48 what does this suggest the actual weight will be of this fruit this summer from the orchards?

6. Let us suppose that the four monthly totals for short stay visitors arriving in a particular country, collected over a four year period, are as shown in the table below

- (a) Calculate the seasonal indices for each of the seasons Jan $\rightarrow$ Apr, May $\rightarrow$ Aug, Sept $\rightarrow$ Dec. (Give indices as percentages correct to 2 dp, eg 105.72%.)
 (b) Deseasonalise the numbers in the arrivals column.

n	Year	Months	Arrivals (millions)
1	1	Jan $\rightarrow$ Apr	1.83
2		May $\rightarrow$ Aug	1.29
3		Sept $\rightarrow$ Dec	1.98
4	2	Jan $\rightarrow$ Apr	1.87
5		May $\rightarrow$ Aug	1.41
6		Sept $\rightarrow$ Dec	2.06
7	3	Jan $\rightarrow$ Apr	1.93
8		May $\rightarrow$ Aug	1.28
9		Sept $\rightarrow$ Dec	2.16
10	4	Jan $\rightarrow$ Apr	2.18
11		May $\rightarrow$ Aug	1.52
12		Sept $\rightarrow$ Dec	2.33

7. A show is held each year over a long weekend, starting on the Friday and finishing on the public holiday Monday. The attendance figures for four consecutive years are shown in the table below.

- (a) Calculate the seasonal index for each of the four days. (Give your answers as a decimal and correct to 4 decimal places, eg 1.1234.)
 (b) Seasonally adjust the numbers in the attendance column. (To nearest 10.)

Year	Day	Attendance
1	Friday	9200
	Saturday	14840
	Sunday	16260
	Monday	16850
2	Friday	9810
	Saturday	15790
	Sunday	17270
	Monday	17800
3	Friday	9450
	Saturday	15220
	Sunday	17200
	Monday	16810
4	Friday	9840
	Saturday	16900
	Sunday	18040
	Monday	16750

8. The table below shows the evening takings for a new take away fish and chip shop, each evening for the first three weeks of operation.

t	Week	Day	Takings	Deseasonalised weekly takings (D).
1	1	Monday	\$537	
2		Tuesday	\$618	
3		Wednesday	\$372	
4		Thursday	\$770	
5		Friday	\$937	
6		Saturday	\$915	
7		Sunday	\$492	
8	2	Monday	\$605	
9		Tuesday	\$730	
10		Wednesday	\$410	
11		Thursday	\$880	
12		Friday	\$1085	
13		Saturday	\$1003	
14		Sunday	\$495	
15	3	Monday	\$705	
16		Tuesday	\$804	
17		Wednesday	\$420	
18		Thursday	\$840	
19		Friday	\$1240	
20		Saturday	\$1270	
21		Sunday	\$531	

- (a) Given that the seasonal indices are as shown below calculate the entries for the deseasonalised weekly takings column, to the nearest dollar.

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
82.42%	96.07%	53.94%	111.87%	145.52%	141.94%	68.24%

- (b) Use a graphic calculator or computer spreadsheet to view a graph showing the deseasonalised figures, D , on the vertical axis plotted against t on the horizontal axis (t is as shown in the above table) to confirm that for the variables t and D linear regression would be appropriate.
- (c) Use t and the deseasonalised data, D , to determine the equation of the regression line $D = A + Bt$.
- (d) Use the regression equation from part (c) and the seasonal indices to predict the takings for Monday, Wednesday and Saturday of week 4.

9. The number of students absent from a school on each day for the first four full weeks of a term are shown in the table below.

n	Week	Day	No. absent	5pt MA (M)	Seasonally adjusted absences
1	One	Mon	35	—	32
2		Tue	25	—	
3		Wed	24	30.4	
4		Thur	28	30.8	
5		Fri	40	31.8	30
6	Two	Mon	37	32.4	34
7		Tue	30	32.8	
8		Wed	27	34.4	
9		Thur	30	36.2	
10		Fri	48	36.6	36
11	Three	Mon	46	37.8	42
12		Tue	32	38.8	
13		Wed	33	40.2	
14		Thur	35	40.6	
15		Fri	55	42.6	42
16	Four	Mon	48	44.0	43
17		Tue	42	45.8	
18		Wed	40	46.0	
19		Thur	44	—	
20		Fri	56	—	42

- (a) Use a graphic calculator, or computer spreadsheet, to view the 5 point moving averages (M) on the vertical axis plotted against n on the horizontal axis to confirm that for the variables n and M , linear regression would be appropriate.
- (b) Use n and the 5 point moving averages, M , to determine the equation of the regression line $M = An + B$.
- (c) Create a completed version of the following to calculate the seasonal indices.

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 1	1.15132				1.31579
Week 2	1.07558				1.39535
Week 3	1.14428				1.36816
Week 4	1.04348				1.21739
Mean (4dp)	1.1037				1.3242

- (d) Calculate the missing values in the final column in the first table
- (e) Use this regression equation $M = An + B$ and the seasonal indices to predict the absences for each day of week five.
- (f) Suggest some real life events that could make these predictions unreliable.

10. Let us suppose that the number of rescues carried out by a surf rescue group in each two months of its six months season, over four seasons are as follows:

n	Season	Months	Rescues	Season's 2 month mean
1	1	Oct/Nov	46	82
2		Dec/Jan	111	
3		Feb/Mar	89	
4	2	Oct/Nov	36	76
5		Dec/Jan	104	
6		Feb/Mar	88	
7	3	Oct/Nov	36	70
8		Dec/Jan	107	
9		Feb/Mar	67	
10	4	Oct/Nov	34	64
11		Dec/Jan	91	
12		Feb/Mar	67	

- (a) Create a completed version of the following:

Rescues as a percent of season's 2 month mean. (As percentages and correct to 2 decimal places).			
	Oct/Nov	Dec/Jan	Feb/Mar
Season 1	56.10%	135.37%	
Season 2			
Season 3			
Season 4			

Seasonal indices (Nearest percent.)	52%		
--	-----	--	--

- (b) Produce deseasonalised figures, D , for each of the two month periods of seasons 1 to 4. (Give the D values to the nearest integer.)
- (c) Use a graphic calculator or computer spreadsheet to view a graph showing the deseasonalised figures, D , on the vertical axis plotted against n on the horizontal axis (n is as shown in the initial table) to confirm that for the variables n and D linear regression would be appropriate. Hence determine the equation of the regression line

$$D = A + Bn.$$

- (d) Use the regression equation from part (b) and the seasonal indices to predict the number of rescues for each two month period of season 5.

Internet web sites for government agencies such as
 The Australian Bureau of Statistics
 and
 The Bureau of Meteorology
 have a wealth of time series data available.

For example

Monthly numbers of house* building commencements approved.

(*Referred to as "dwelling units".)

Quarterly consumer sales figures.

Monthly maximum temperatures.

Etc.

Investigate.

Miscellaneous Exercise Two.

This miscellaneous exercise may include questions involving the work of this chapter, the work of any previous chapters, and the ideas mentioned in the preliminary work section at the beginning of the book.

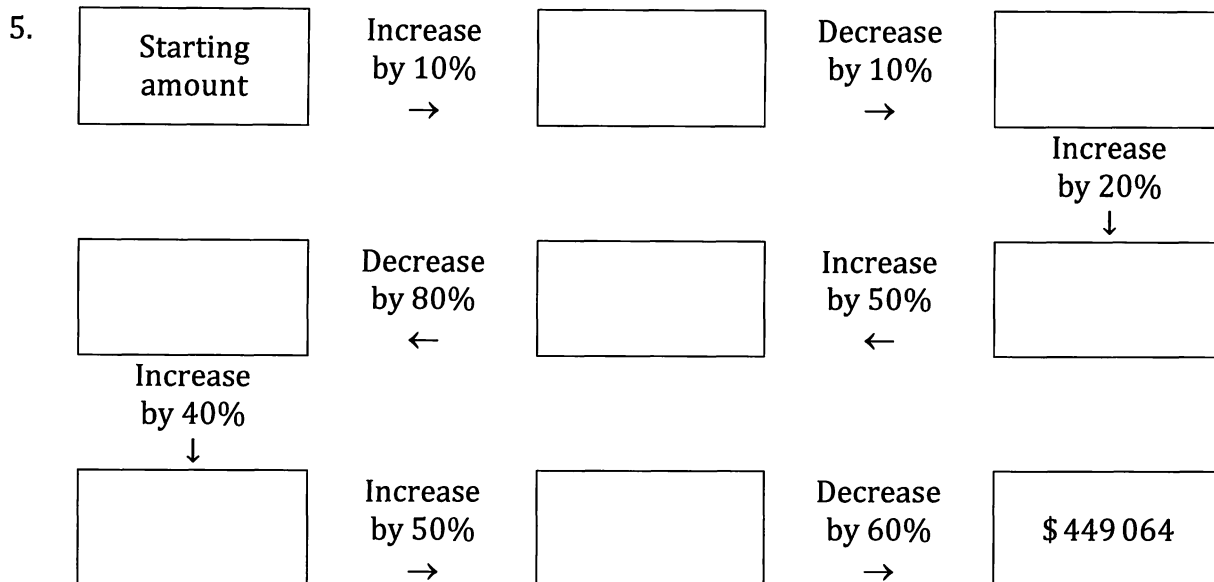
1. Increase \$250 by 10% then decrease your answer by 10%.
2. The rule for converting a Celsius temperature ($^{\circ}\text{C}$) to the corresponding Fahrenheit temperature ($^{\circ}\text{F}$) is

$$F = \frac{9}{5}C + 32.$$

What is the value of the correlation coefficient r_{CF} ?

3. What does it mean if the seasonal index for February is 0.87 (or 87%).
4. Deseasonalise the following raw data given the seasonal indices are as stated.

Season	Spring	Summer	Autumn	Winter
Raw data	1400	1940	1520	750
Seasonal index	1.40	1.25	0.95	0.4



Find the starting amount.

6. Counting the year 1998 as zero, the year of manufacture and the asking price of a certain make and model of vehicle advertised in a newspaper one day in 2014 were as follows:

Year N ^o .	5	13	6	4	4	15	6
Asking price (\$)	19980	44500	17000	13500	11400	57300	20500
Year N ^o .	11	10	7	12	10	6	
Asking price (\$)	39000	32900	25000	35700	36800	25500	

Display these figures graphically with t , the year number in the above table, on the horizontal axis and p , the asking price, on the vertical axis.

Comment on any relationships shown on your graph.

Use your calculator or a spreadsheet to determine the equation of the line of best fit in the form $p = at + b$ and use this to predict the likely price in 2014 for a 2007 vehicle of this make and model.

7. Using sales data from previous years a company determines that the quarterly revenues achieved from sales of its products seem to follow a reasonably steady seasonal pattern with the following seasonal indices:

1st quarter	0.85	2nd quarter	1.25
3rd quarter	1.12	4th quarter	0.78

The following year the company finds that the quarterly revenue figures are as follows:

1st quarter	\$752 000	2nd quarter	\$1 138 000
3rd quarter	\$1 176 000	4th quarter	\$694 000

- (a) If the above information is all we know about the company why are the quarterly revenues shown above a surprise?
- (b) If we were informed that the company had a big advertising campaign at the start of the 3rd quarter might this make the revenue figures less surprising?

8. The table below shows the sales figures for a particular item on a quarter year basis, for three years.

(a) Create completed versions of the following tables using the average percentage method to determine seasonal indices.

t	Year	Quarter	Number sold in quarter.	Number sold as percentage of quarterly mean for the year.	Number sold seasonally adjusted.
1	One	1st	1637	106.99%	1514
2		2nd	1489		
3		3rd	1244		
4		4th	1750		
5	Two	1st	1405		
6		2nd	1241		
7		3rd	1012		
8		4th	1538		
9	Three	1st	1253		
10		2nd	1121		
11		3rd	852		
12		4th	1362		

	Year 1	Year 2	Year 3	Seasonal index
1st Quarter	106.99%			108.13%
2nd Quarter				
3rd Quarter				
4th Quarter				

- (b) Use a graphic calculator or computer to view the graph of the seasonally adjusted figures, which we will call S , plotted on the vertical axis, against t on the horizontal axis, and confirm that applying linear regression techniques is appropriate.
- (c) Use t and the seasonally adjusted sales figures to determine the equation of the regression line $S = at + b$.
- (d) Use this regression equation and the seasonal indices to predict the number of the particular item sold for each quarter of year 4.